Name:	BUID:

WRITING AND ARGUMENT ORGANIZATION WORKSHOP

Communication of results is the final step in the scientific method – making sure that the work you've done in your lab is available to others for their reference, their comment, and their use. While none of the experiments we will do in this course will be novel enough to merit true communication in the literature, it is important to begin developing these skills and tools so that you will be prepared when the time does come to engage in this type of scholarly work.

You will **not be writing a scholarly paper** (research paper/journal article) in this assignment. Instead, this workshop will focus on building and developing the skills necessary to effectively and efficiently formulate and organize a cogent, logical, and successful argument. In reality, this task is the most *intellectually challenging* part of writing scholarly papers, and once it is complete the remaining writing process can be rather expedient – the logical flow and most of the language will already be decided in your checklist, and all that will remain will be to expand the checklist into flowing prose. To that end, we will focus on three skills that are critical in this process: (1) making broad, meaningful claims; (2) planning and researching logical arguments (and diagramming them using Graphical Representations of Argument, GRAs); and (3) preparing a detailed checklist (outline) that can be expanded into a full, scholarly paper.

Teamwork in science

An important thing to realize early in your career is that doing science is a team sport – individuals operating separately but collectively helping each other toward individual, mutual, and communal goals. Academic research groups meet regularly, often weekly, to discuss the research being done and to help each other move forward with their work – this is usually called "group meeting." Group meetings can range from the formal to the very informal, and can often include dozens of individuals ("supergroup" meetings). At these meetings, principle investigators, post-docs, graduate students, and undergraduate students share their work, ask for feedback, defend their findings, and grow in their understanding.

In that spirit of group collaboration, the first two parts of the workshop will be done in groups of 3 (or 4) that will be assigned by your instructor. Each group will be assigned to one of the following experiments that you've completed to work with (there may be a couple of groups in each section working with the same lab):

- Lab #2: molecular size determination
- Lab #5: modeling conjugated dyes with the multielectron particle-in-a-box
- Lab #7: colorimetric determination of iron content

The final part of the workshop – developing a detailed checklist for the paper – will be done individually. All of the work should be completed on this handout and submitted to Gradescope before the end of the lab session.

Group members: ______________________ ______________________

______________________ ______________________

Name:	BUID:

Getting started – making a claim (in groups)

Background reading

Before starting, take a few minutes to read through chapter 1 of the Undergraduate's Guide to Writing in the Sciences (posted on the course Blackboard site); this chapter will give you important context to the type of writing that you will do in the sciences and in this course. Then, review sections 5.1 and 5.2, and focus on the concept of *Claims* and *Grounds/Evidence* – these will be the topics of this first section.

Initial claim

After completely analyzing the data collected in an experiment, the first stage in preparing to write a paper is to clarify the claim that you intend to make. Work with your group to settle on a strong claim – the most broad claim that you feel can be adequately supported by the data. You will be asked to report back your group's claim once all of the groups have settled on a claim.

Revised claim

After hearing from the all of other groups, and the discussion that followed, propose an updated claim for the lab you were assigned. You will be asked to report back your group's revised claim once all of the groups have finished.

Name:	BUID:

Planning a strong argument (in groups)

Background reading

Read the remainder of chapter 5 in the writing guide. Focus on (a) the components that go into making a strong argument, and (b) how GRAs can be used effectively to plan the paper that you will eventually write.

Preparing a Graphical Representation of Argument

After completely analyzing the data collected in an experiment, the first stage in preparing to write a paper is to clarify the claim that you intend to make and prepare a logical argument based on this claim. As discussed in chapter 5 of the Undergraduate's Guide to Writing in the Sciences, many components are necessary to produce a successful argument (warrants, backing, etc.), all of which must be researched and identified before you start writing. Often, the most time-consuming part of forming an argument is the research that needs to be done to make sense of the results and support your argument components (warrants, backing, etc.). You will likely find a lot of overlap between the answers to the post-lab "questions for thought" and the research needed to prepare a strong argument.

Once you've collected these necessary components, the most effective way to plan your argument (read: paper) is to assemble them into a logical argument using a Graphical Representation of Argument (GRA). Your GRA is then used to effectively outline the paper that you will eventually write. This is the subject of this section: preparing a GRA for your group's assigned lab that will lead to a strong, well-reasoned argument.

Work in your group to prepare a good GRA for your assigned lab. Use the rest of this page to start listing the components of the argument that you could conceive of including, and work on the GRA on the next page. Make sure to consult with your LA and TF to get feedback and guidance on your GRA – remember, team sport! Only continue to the next section once your group, your TF, and your LA are satisfied with your GRA.

Name:	BUID:

(Diagram your GRA here)

Name:	BUID:

Preparing a detailed checklist (individual work)

Background reading

Chapter 6 of the Undergraduate's Guide to Writing in the Sciences covers the sections included in an abridged paper (i.e., Title, Results and Discussion, Conclusion, References). Sections 6.5 and 6.6 discuss how checklists are used to organize and prepare a strong argument. In particular, section 6.5 discusses how experts approach the task of paper-writing and use checklists to streamline their work; the sample checklist at the end of the chapter is a good example of what you will produce by the end of this assignment.

Developing a GRA into a checklist for a strong paper

In the previous section we worked on preparing a GRA that represents the complete logical argument that you intend to make in the paper. In this final section we will take the next step toward writing the paper: preparing a detailed checklist (outline) for the abridged paper based on the GRA. Your main objective for this section will be to produce a clear, **detailed** checklist that could be used to write a strong paper for your assigned lab. Details about checklists and their structure are given in chapter 6 of the writing guide. Pay special attention to the sample checklist that is presented at the end of that chapter – this is a good model for what you will produce. See section 6.5.2 for detailed instructions about how to go from your completed experiment to making a strong checklist.

For each component of your checklist that is based on research and reading that you've done, include a citation following the format in chapter 3 of the writing guide. Prepare and include a References section (following the guidelines in chapter 3 of the writing guide) for your checklist.

Please note: in CH109 we will focus on writing *abridged papers*, which are comprised of a Title, Results and Discussion, Conclusion, and References (i.e., no Introduction or Experimental sections). In the writing guide, chapter 6 discusses all of the sections included in the abridged paper, except for References (chapter 3). You will be writing an abridged paper on one of the labs assigned in this workshop; these sections will guide you in arranging the components from the GRA (and the objective, which is not in your GRA) that need to be included in your checklists and eventually your paper!

While each student will prepare their own checklist, you are encouraged to consult with your TF and LA for guidance as you work. Also, after you have finished your work, do feel free to compare your work with other students and get their feedback. The goal is that you produce a good final product.

Submitting this workshop

When you are done with your checklist (or at the end of lab), scan and upload this worksheet to **Gradescope**.

Continue writing your checklist if it's not finished - while you won't submit it, it can be an outline for writing your paper on the experiment!